GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Cold Bends**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-007
Document Revision	R01
Document Status	Issued for Review
Originator	Emeka Nwachukwu
Security Classification	Restricted
Issue Date	12-Jul-22

Specification for Cold Bends

R01	12.07.22	Issued for Review	E. Nwachukwu	M. Ekpunobi	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	12-07-22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
   for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-007

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipes
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006	Data Sheets for Induction Bends
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
DEP 31.40.00.10-Gen	Pipeline Engineering (Amendments/Supplements To ISO 13623)
DEP. 82.00.10.10-Gen	Project Quality Assurance
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
MPS.	Manufacturing Procedure Specification
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

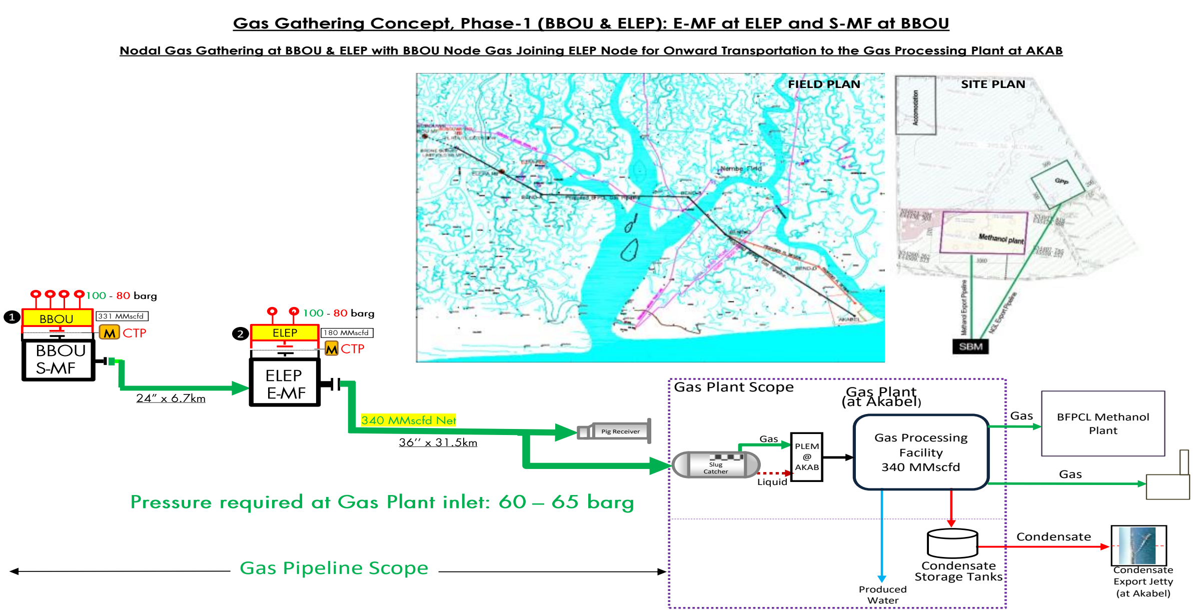

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This Specification covers the minimum requirement for the fabrication of field/cold bends for used to achieve connectivity for phase 1 of BFPCL Gas Gathering Pipelines using onsite field bending machine.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

2.1. International Codes and Standards.

Document Number	Document Title
API 5L	Specification for Line Pipe
API 1104	Welding of Pipelines and Related Facilities
ASME B31.8	Gas Transmission and Distribution Piping Systems
BS 8010-2.8	Code of Practice for Pipelines. Pipelines on land: design, construction land installation.
ISO 13623	Petroleum and Natural Gas Industries – Pipeline Transportation Systems

2.2. Project References

Document Number	Document Title
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipes
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009	Pipeline Construction Specification
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-001	Data Sheet for Line Pipes
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-007	Data Sheet for Cold Bends

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- COMPANY's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the COMPANY for resolution.

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	**6.7km**
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L–X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	24"
Ultimate Tensile Strength	570 MPa
Wall Thickness	**15.88mm**
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

3.2. 36” Multiphase Pipeline Design Data

The 36” pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	**31.5km**
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipe-	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L–X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield	483 Mpa
NPS	36"
Ultimate Tensile Strength	570MPa
Wall Thickness	**23.83mm**
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coat-	3LPE
Cathodic Protection	Impressed current

4. General Requirements

Cold bends shall be made by the cold stretch smooth bending method using bending machines equipped with internal padded mandrel, padded bending shoes and drive rollers. Prior to construction, the equipment shall be tested and qualified, witnessed by COMPANY's representative, for cold bending to verify that the pipes shall be bent in accordance with applicable codes and this specification. The bends shall have uniform bearing under the pipe and will maintain the required depth of cover and shall be neither in tension nor in compression when placed in trench.

For fabrication of the field cold bends, refer to EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-007 Data Sheet for Cold Bends

4.1. Pipe

- The materials to be used in the fabrication of the cold bends shall be the mother pipes for the pipeline right of way construction with all the certifications and documentations confirmed by the cold bending inspectors and the COMPANY representatives.

4.2. Bend Wall Thickness

- The wall thickness of the bends due to stretching shall not be less 95% of the starting wall thickness or less than the nominal wall thickness minus the manufacturing tolerance which ever one that is greater. Reverse, mitred or wrinkled bends shall not be permitted. Any pipe bends showing evidence of buckling or wrinkling shall be cut out and replaced. No combination of bends shall be allowed in a single joint of pipe.

4.3. Tangent Length

- All bends shall have a minimum tangent length of 500 mm, unless otherwise specified by COMPANY.

4.4. Bend Radius

- The minimum radius of field bends shall not be less than 40 times the outside pipe diameter. Pipes shall be bent with a uniform radius and the curvature shall be distributed in such a way that a permanent deviation not greater than 1.5 degree is achieved on each pipe section of a length equal to the diameter of the pipe. The straight ends of each bent length shall not suffer deformations due to bending.

4.5. Bend Ovality

- The Ovality of pipe cross section after bending shall not exceed 2.5 percent. Check shall be made during Pipeline quality tests, to ensure that the bend shall pass the specified gauging plates. Gauging plates assembly shall consist of two plates with the external diameter equal to 96% of the internal pipe diameter (ID), spaced at one pipe diameter apart.

4.6. Pipe Coating Dis-bonding.

- Bending will be accomplished without damaging or disbonding the coating to the pipe. Any damage to the pipe coating due to bending shall be repaired in accordance with the requirements of Specification and the repair procedure approved by COMPANY.

4.7. Other requirements

- CONTRACTOR shall make in the field all pipe cold bends required for the construction of the pipeline in accordance with the requirements for minimum cover, vertical profiles and horizontal alignment.
- Cold bending is to be carried out after the trench has been dug, except in special cases, where approval of the COMPANY Representative shall be sought.
- CONTRACTOR shall furnish the qualified engineers and assistant necessary to determining location and sizing of all bends. All field bends shall be made by the cold stretch smooth bending method using bending machines equipped with internal padded mandrel, padded bending shoes and drive rollers.
- CONTRACTOR shall ensure that all bends are free from wrinkles or buckles with the minimum of stretching. Reverse, mitred or wrinkled bends shall not be permitted. Any pipe bends showing evidence of buckling or wrinkling shall be cut out and replaced.
- The bends, when in place in the trench, shall have uniform bearing under the pipe and will maintain the required depth of cover indicated on the Construction Drawings and, when in place in the trench, shall be neither in tension nor in compression.
- Special attention shall be paid to bending requirements at valve and crossing sites, where specific pipeline profile must be observed.
- The bending procedure shall be developed by CONTRACTOR and submitted to COMPANY for approval.
- Prior to construction, CONTRACTOR shall test the equipment for cold bending to verify that the bends shall be performed as specified by the applicable codes and this Specification.
- No combination of bends shall be allowed in a single joint of pipe. Sag and over-bends are not considered combinations since both are on the same axis. Under no circumstances shall heat be applied to pipe for the purpose of bending.
- The longitudinal welded pipe to be bent shall have the seam weld located in the neutral axis of the bend. For the horizontal bends, the longitudinal weld shall be positioned in the upper

part. If horizontal deviations must be achieved by consecutive bending of pipes, the bending of the pipes shall be made by positioning the longitudinal weld alternatively on the right and on the left of the plane passing through the neutral axis defined above, in such a way that the single bends are welded with the longitudinal welds staggered and separated 30 degrees from neutral axis. When vertical deviations are to be achieved by consecutive bends, the longitudinal welds shall be diametrically opposed.

- The locations of field cold bends are shown on alignment sheets and detailed crossing drawings.